📋 Quiz 1 — Study Guide & Syllabus

> **Course:** RDBMS Lab | **Quiz Date:** 13/04/2026
> **Marks:** 20 | **Duration:** 40 minutes
> **Covers:** Lab 01, Lab 02, Lab 03, Lab 04

📢 General Instructions

- This is a **closed-book, no-internet** exam.
- The quiz uses a **scenario-based** database. The schema and data will be provided in the question paper — **you do not need to memorize any tables**.
- Focus on **understanding concepts** and **writing correct SQL queries**.
- Partial marks are given for correct logic even with minor syntax errors.
- **No marks deducted** for missing semicolons.

📚 Topics to Study

From Lab 01 — Introduction to RDBMS

Topic	What You Should Know
What is RDBMS	Tables, rows, columns, PK, FK
ACID Properties	Define each: Atomicity, Consistency, Isolation, Durability
SQL Categories	Know which commands belong to DDL, DML, DCL, TCL
Data Types	When to use INT, VARCHAR, DECIMAL, DATE, ENUM, BOOLEAN
Constraints	PK, NOT NULL, UNIQUE, DEFAULT, CHECK, AUTO_INCREMENT
CREATE TABLE	Write correct syntax with columns and constraints
INSERT INTO	Single-row and multi-row insert syntax

Practice:

- Write a `CREATE TABLE` from scratch with at least 5 constraints.
- Identify what SQL category a command belongs to (e.g., `ALTER TABLE` → DDL).

From Lab 02 — Database Design

Topic	What You Should Know
Normalization	1NF: atomic values. 2NF: no partial dependency. 3NF: no transitive dependency

Topic	What You Should Know
When a table violates 1NF/2NF/3NF	Given a table, identify which normal form is violated and why
Primary Key types	Surrogate (AUTO_INCREMENT) vs Natural (email) vs Composite
Foreign Keys	Syntax, purpose, and what happens with ON DELETE CASCADE / SET NULL / RESTRICT
DROP vs TRUNCATE vs DELETE	Know the differences: structure removal, WHERE clause, recoverability, AUTO_INCREMENT
ALTER TABLE	ADD, MODIFY, RENAME, DROP columns

Practice:

- Given an unnormalized table, convert it to 3NF step by step.
- Explain what happens when you delete a parent row with CASCADE vs SET NULL.

From Lab 03 — Basic SQL Queries

Topic	What You Should Know
SELECT + WHERE	Filtering rows with conditions
Logical operators	AND, OR, NOT — and why **parentheses matter**
BETWEEN, IN, LIKE	Range checks, list matching, pattern matching (`%` and _)
String functions	CONCAT, UPPER, LOWER, LENGTH, SUBSTRING, REPLACE
Date functions	NOW, CURDATE, YEAR, MONTH, DATEDIFF, DATE_FORMAT
NULL handling	`IS NULL`, `IS NOT NULL`, why `= NULL` doesn't work, COALESCE, IFNULL
NULL arithmetic	`10 + NULL = NULL`, `NULL = NULL` is **not** TRUE
ORDER BY	ASC/DESC, multi-column sorting
LIMIT / OFFSET	Top-N queries, pagination
DISTINCT	Removing duplicate rows from results
Aliases	Column aliases (`AS`), table aliases for JOINs

Practice:

- Write a query using LIKE with `%` to find emails from Gmail.
- Use COALESCE to replace NULL with a default value.
- Predict the output of `SELECT NULL + 5, NULL = NULL, NULL IS NULL;`

From Lab 04 — Intermediate SQL (JOINs, Aggregates, Subqueries)

Topic	What You Should Know
INNER JOIN	Returns only matching rows from both tables
LEFT JOIN	All from left table + matching from right (NULL if no match)
RIGHT JOIN	All from right table + matching from left
LEFT JOIN + WHERE IS NULL	The "find missing / orphan" pattern
Multi-table JOINs	Chaining 3+ tables in one query
COUNT, SUM, AVG, MIN, MAX	Aggregate functions — collapse rows into a single value
GROUP BY	Split rows into groups, apply aggregate per group
HAVING	Filter **groups** (vs WHERE which filters **rows**)
WHERE vs HAVING	WHERE runs before GROUP BY; HAVING runs after. Aggregates only in HAVING.
SQL Execution Order	FROM → WHERE → GROUP BY → HAVING → SELECT → ORDER BY
Subqueries	Scalar (single value), table (IN list), correlated (references outer query)
EXISTS vs IN	EXISTS stops at first match; IN evaluates entire subquery

⚡ **This is the most heavily tested topic. Practice writing queries with JOINs + GROUP BY + HAVING.**

Practice:

- Find customers who haven't placed any orders (LEFT JOIN + IS NULL).
- Count products per category, show only categories with > 3 products.
- Find products priced above the average (subquery in WHERE).
- Write a 3-table JOIN with GROUP BY and HAVING.

📐 Quiz Format (what to expect)

Section	Type	Marks	Time Estimate
A	5 Multiple Choice Questions	5	~8 min
B	2 questions: Predict query output / Find an error	3	~8 min
C	6 SQL query writing questions (scenario-based)	12	~22 min
Total		**20**	**~38 min**

💡 Tips for Success

1. **Read the schema and data carefully** before writing any query.
2. **Start with the easy questions** in Section C — don't get stuck on hard ones.

3. For JOINs: always ask yourself *"Do I need ALL rows from one side?"* → if yes, use LEFT JOIN.
4. For GROUP BY: if the question says "per category" or "per customer" → that's GROUP BY.
5. For HAVING: if the question says "only show groups where count > X" → that's HAVING.
6. **Write column aliases** (`AS`) — it makes your queries clearer and shows understanding.
7. Practice the **"find missing" pattern**: `LEFT JOIN ... WHERE right_table.pk IS NULL`.

📖 Study Resources

- Lab notes: `LAB_01_NOTES.md` through `LAB_04_NOTES.md`
- Example SQL files in each lab's `examples/` folder
- Practice by running queries on `ecommerce_db`

> **Good luck!** 🍀